Bayesian Simple Normal Regression

DAEN 427 - Decision and Risk Analysis

Alexander Abuabara

Data Engineering, Texas A&M University

Fall 2026

Learning objectives

- Understand the data analysis workflow.
- Write concise Python and R examples.
- Compare outputs across languages.

Python | A simple example

A short Python example using a list and its mean.

PYTHON

```python
values = [4, 8, 12, 16]
mean_value = sum(values) / len(values)
print(f"Mean: {mean_value:.1f}")
```

Expected output: `Mean: 10.0`

R | The same calculation

The equivalent operation in base R.

```r
values <- c(4, 8, 12, 16)
mean_value <- mean(values)
print(mean_value)
```

Expected output: `[1] 10`

Python and R | Quick comparison

Task	Python	R
Create vector	`[1, 2, 3]`	`c(1, 2, 3)`
Length	`len(x)`	`length(x)`
Mean	`sum(x)/len(x)`	`mean(x)`

Key takeaways

- Use short, readable examples on each slide.
- Keep slides visually consistent.
- Prefer one idea per slide.

Practice

Question:

A. Réponse A

B. Réponse B

C. Réponse C (bonne réponse)

D. Réponse D

Practice

Question:

A. Réponse A

B. Réponse B

C. Réponse C (bonne réponse)

D. Réponse D